ARCHIVES OF BIOCHEMISTRY AND BIOPHYSICS
Vol. 204, No. 1, October 1, pp. 350–360, 1980

The Effects of Calcium and Lanthanide Ions on the Activity of Bovine Heart Nicotinamide Adenine Dinucleotide-Specific Isocitrate Dehydrogenase[1]

TADASHI AOGAICHI, JANINE EVANS, JEROME GABRIEL, AND GERHARD W. E. PLAUT

Department of Biochemistry, Temple University School of Medicine, Philadelphia, Pennsylvania 19140

Received April 24, 1980

(i) The activity of purified NAD-specific isocitrate dehydrogenase from bovine heart was stimulated by free Ca^{2+} in the presence of ADP and subsaturating levels of magnesium isocitrate, but not in absence of ADP. However, Ca^{2+} was not absolutely required for ADP activation. This was particularly apparent when free Mg^{2+} was kept low (0.0024–0.020 mM) and the substrate magnesium DL-isocitrate ranged from 0.07–0.25 mM. When kinetic constants were determined at pH 7.4 under these conditions and in the absence of ethylene glycol bis(β-aminoethyl ether) N,N'-tetraacetate, Ca^{2+} had little or no effect on K_m (app) for ADP; the stimulation of rate by Ca^{2+} was mainly due to increased V (app). With subsaturating ADP, there was an interdependence in the interaction of the enzyme with substrate and Ca^{2+}. Thus, with ADP constant (0.30 mM) the values of K_m (app) for magnesium DL-isocitrate declined from 0.35 mM at zero Ca^{2+} to 0.19 mM with saturating Ca^{2+} without affecting V; K_m (app) for free Ca^{2+} declined with increasing magnesium isocitrate to a limiting K_m of 0.3 μM. (ii) Ethylene glycol bis(β-aminoethyl ether)-N,N'-tetraacetate, frequently used as a calcium buffer, inhibited enzyme activity with and without ADP. (iii) The enzyme was not inhibited by the calmodulin inhibitors trifluoperazine and chlorpromazine. Inhibition by lanthanide ions of the isocitrate dehydrogenase was competitive with magnesium isocitrate and not with respect to Ca^{2+}. The values of K_{is} (1.8 to 3.1 μM) for La^{3+}, Yb^{3+}, Gd^{3+}, Eu^{3+}, Tb^{3+}, and Er^{3+} were about two orders of magnitude smaller than K_m for magnesium DL-isocitrate.

NAD-specific isocitrate dehydrogenase (EC 1.1.1.41) from a number of mammalian tissues is modulated positively by ADP and inhibited by NADH, NADPH, and ATP. The enzyme is active without ADP and activation by the nucleotide above pH 6.8 is due to lowering of the K_m of substrate without change of V (1, 2). There is an interdependence in the interaction of the enzyme with the substrate magnesium isocitrate and the modifier ADP (3). NAD-isocitrate dehydrogenases from a variety of animal tissues have been shown to require Mg^{2+}, Mn^{2+} or Co^{2+} for activity (2, 4–6). Since the enzymes from yeast (7) and heart (8) are inhibited by cyanide the metal content of the homogeneous enzyme from bovine heart was examined. However, atomic absorption spectrophotometry indicated less than 1 g atom per 300,000 g of protein of Fe, Cu, Zn, Mo, Ni, Co, Mn, Cr, Ca, and Mg (9).

Both activation and inhibition of NAD-specific isocitrate dehydrogenase activity in extracts of animal tissues by added calcium have been reported. Vaughan and Newsholme (10) noted inhibition by calcium of the enzyme activity from rat heart. In contrast, Zammit and Newsholme (11) found that increasing calcium concentration (up to 10 μM) had no effect on the dehydrogenase from kidney cortex and brain extracts either with or without ADP, and they attributed physiological significance to these tissue variations in response to calcium. In extracts of muscles from the whelk, locust, and waterbug, increase in calcium concentration inhibited enzyme

[1] Supported in part by Research Grant AM 15404 from the National Institute of Arthritis, Metabolic and Digestive Diseases, U. S. Public Health Service.

0003-9861/80/110350-11$02.00/0

activity at low substrate concentrations when ADP was absent whereas, in the presence of ADP, dehydrogenase activity was increased by Ca^{2+} (11). Values of K_m for isocitrate, magnesium, and ADP were increased upon addition of Ca^{2+} with the partially purified enzyme from blowfly flight muscle mitochondria; activation of the enzyme by Ca^{2+} could not be confirmed (12, 13). However, Denton *et al.* (14) reported that calcium or strontium activated NAD-isocitrate dehydrogenase from extracts of rat heart mitochondria by lowering the K_m (app) for isocitrate without affecting V in the presence of ADP; they observed a slight increase in K_m for isocitrate when ADP was absent.

Variations in reported results on the effect of calcium may be due to the precision of determinations of the ionic species in the reaction mixtures (12) or due to differences in kinetic properties of enzymes from the various animal species examined (e.g., insect flight muscle vs rat heart). However, it is possible that other enzymes in extracts from animal tissues may utilize isocitrate and complicate the kinetic analyses of isocitrate dehydrogenase activity. For example, England *et al.* (15) found that Mg^{2+} and Ca^{2+} have a marked effect on the equilibrium citrate/isocitrate concentration for aconitase at pH 7.3, due to the different affinities of these anions for the bivalent cations. In order to avoid this complication, in the present studies the effects of calcium were examined with purified homogenous NAD-specific isocitrate dehydrogenase from bovine heart. EGTA,[2] which was used as a calcium buffer in all of the above investigations (10–14), was found to be an inhibitor of the bovine heart enzyme under certain conditions. The kinetic constants for magnesium isocitrate, ADP, and Ca^{2+} reported here were therefore determined in the absence of EGTA. In addition, we have investigated the effects on NAD-isocitrate dehydrogenase of a number of organic (phenothiazine derivatives) and inorganic (lanthanides) substances which have been reported to affect other calcium-dependent reactions.

[2] Abbreviations used: Hepes, 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid; EGTA, ethylene glycol bis(β-aminoethyl ether)N,N'-tetraacetic acid; DHPTA, 1,3-diamino-2-hydroxypropane-N,N,N',N'-tetraacetic acid.

EXPERIMENTAL PROCEDURES

Materials. ADP, NAD^+, and EGTA were purchased from Sigma Chemical Company. Hepes from Calbiochem was recrystallized from ethanol–water. DHPTA and DL-isocitric lactone were from Aldrich. DL-Isocitric lactone was recrystallized from ethyl butyrate and hydrolyzed with alkali to isocitrate before use. $LaCl_3$, $Yb(NO_3)_3$, $Gd(NO_3)_3$, $EuCl_3$, $TbCl_3$, and $ErCl_3$ were from Alfa-Ventron; other inorganic salts of reagent grade were from Mallinckrodt. Trifluoperazine was kindly donated by Dr. V. D. Wiebelhaus of Smith, Kline and French Laboratories.

All reagents were dissolved in redistilled deionized water and stored in Teflon vessels. Pipets and other glassware were washed in sulfuric acid–nitric acid cleaning solution and rinsed in deionized water.

NAD-specific isocitrate dehydrogenase from bovine heart prepared by the method of Giorgio *et al.* (16) had a specific activity of 28 μmol of NAD^+ reduced/mg of protein/min at 25°C under the conditions of assay reported previously (17).

Assays. Reaction mixtures contained 167 mM Na-Hepes at pH 7.4, 0.5 mM NAD^+, and enzyme. DL-Isocitrate, Mg^{2+}, Ca^{2+}, ADP, chelators, or inhibitors were added at the concentrations indicated in the text and in table and figure legends. Reactions were initiated by the addition of NAD^+ and activity was determined by measuring initial velocities at 340 nm at 25°C. Deviations from these incubation conditions have been noted in the text.

The total calcium content of assay mixtures and stock solutions was assayed by atomic absorption spectrometry.

Apparent stability constants at pH 7.4. The constants (mM^{-1}) were calculated from stability constants and pK values reported in the literature. The values used, as shown in parentheses, were as follows: magnesium isocitrate$^-$ (0.525) and calcium isocitrate$^-$ (0.457) (18), magnesium ADP$^-$ (1.36), calcium-ADP$^-$ (0.365), magnesium EGTA^{2-} (0.048), calcium EGTA^{2-} (29,840), magnesium DHPTA^{2-} (0.141), and calcium DHPTA^{2-} (9.50) (19).

Calculations. The concentrations in reaction mixtures of complexes of isocitrate, ADP, and chelators with calcium and magnesium were calculated from the apparent stability constants at pH 7.4 with the aid of the computer program of Feldman *et al.* (20). Kinetic parameters of the enzyme were calculated from initial velocities by fitting the data to the appropriate computer programs developed by Cleland (21) and as described previously for NAD-specific isocitrate dehydrogenase (3).

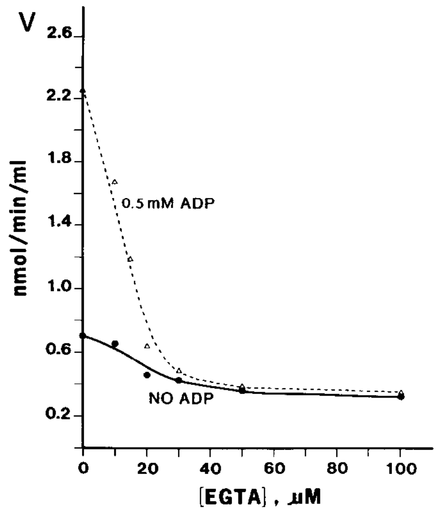

FIG. 1. The effect of varying EGTA at high magnesium: isocitrate ratio. Reaction mixtures contained 1.0 mM $MgSO_4$ and 0.20 mM DL-isocitrate. When present the concentration of ADP was 0.50 mM. EGTA was varied from 0 to 100 μM. The endogenous calcium content was 2.3 μM.

RESULTS AND DISCUSSION

Effect of Calcium

Experiments with high magnesium: isocitrate ratios. In the initial experiments the conditions of incubation were similar to those of Denton *et al.* (14), i.e., the concentration ratios of total magnesium to total isocitrate were kept relatively high (magnesium: isocitrate from 5:1 to 1.7:1). Denton *et al.* (14) conducted the reactions with rat heart extracts at pH 7.0; the incubations here were done at pH 7.4, since the stimulation of activity by ADP of the bovine heart enzyme was larger at pH 7.4 than at pH 7.0 (22).

Under the above experimental conditions, in confirmation of Denton *et al.* (14), the activation of enzyme activity by ADP was almost completely suppressed by EGTA when examined either as a function of EGTA concentration with ADP constant (Fig. 1), or when the effect of varying ADP was compared in absence or presence of 100 μM EGTA (Fig. 2). Furthermore, in the presence of ADP (0.50 mM), the addition of increasing Ca^{2+} caused progressive reversal of inhibition by EGTA (1.0 mM); the reaction rate became nearly maximal above 1.3 μM free Ca^{2+} (Fig. 3).

It is uncertain from the above results whether the enhancement of reaction rate caused by Ca^{2+} in the presence of ADP was due to lowering of K_m for ADP or due to increased V. The resolution of this problem is complicated by the substantial inhibition of activity by EGTA. When ADP was absent, the addition of 100 μM EGTA inhibited the reaction rate by about 73% (Fig. 2); inhibition was already evident between 10 and 20 μM EGTA, maximal between 50 and 100 μM EGTA (Fig. 1), and did not increase further when EGTA was raised to 1000 μM (not shown). The decrease in activity due to EGTA addition could not be attributed to a decline in substrate concentration; in the experiment shown in Fig. 2, when ADP was absent, the concentrations of magnesium isocitrate (calculated from the stability constants) were 0.0658 mM without EGTA and 0.0656 mM with 100 μM EGTA. Without ADP, the inhibition by EGTA could not be reversed by the addition of calcium (Fig. 3). Addition of Ca^{2+} did not significantly inhibit, either with or without EGTA, as long as the calcium levels were not high enough to appreciably lower magnesium isocitrate by competitive chelation by Ca^{2+} of available isocitrate. The lack of significant inhibition by about 7 μM *free* Ca^{2+} in the presence of EGTA (1000 μM) and the absence of ADP

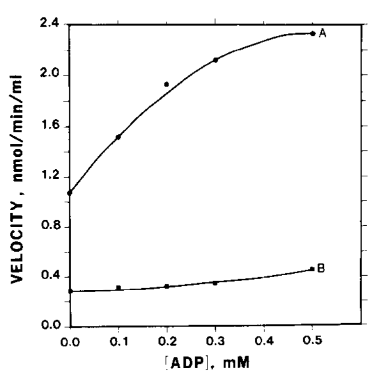

FIG. 2. The effect of EGTA on ADP activation at high magnesium: isocitrate ratio. Reaction mixtures contained 1.0 mM $MgSO_4$ and 0.20 mM DL-isocitrate. The endogenous calcium content was 3.0 μM. ADP was varied from 0.00 to 0.50 mM. (A) Without EGTA; (B) with 100 μM EGTA.

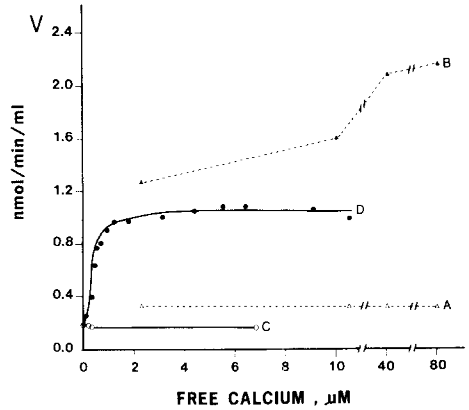

FIG. 3. Comparison of effects of Ca^{2+} with and without ADP and with and without EGTA. Reaction mixtures contained 1.0 mM $MgSO_4$ and 0.60 mM DL-isocitrate maintaining magnesium isocitrate between 0.14 and 0.15 mM. The concentration of endogenous calcium was 3 μM; Ca^{2+} was varied by the addition of $CaCl_2$ and the final concentrations of free Ca^{2+} were calculated from the stability constants. Further additions were as follows: (A) none; (B) 0.50 mM ADP; (C) 1.0 mM EGTA; (D) 1.0 mM EGTA and 0.50 mM ADP.

(Fig. 3) also indicated that Ca-EGTA was not an inhibitor when magnesium isocitrate (0.15 mM) was maintained with a relatively high ratio of total magnesium: isocittrate (1.7:1).

EGTA also affected activity in the presence of ADP by mechanisms in addition to lowering the concentration of activating Ca^{2+}. This became particularly apparent when the effects of increasing Ca^{2+} were examined with constant ADP (0.50 mM) in the absence and presence of EGTA as shown in Fig. 3. With EGTA velocity was maximal between about 2 and 11 μM free Ca^{2+}; however, within the same range of Ca^{2+} concentrations, activity was significantly higher without than with EGTA.

Experiments with high isocitrate: magnesium ratios. To diminish the possible complications of the inhibition of NAD-isocitrate dehydrogenase by free Mg^{2+} (3), the effects of Ca^{2+} on ADP activation were examined under conditions where magnesium isocitrate (range 0.07 to 0.25 mM) was maintained at high total isocitrate: magnesium ratios (740:1 to 90:1), yielding concentrations of free Mg^{2+} ranging from 0.0024 to 0.020 mM. When examined at two substrate concentrations (0.065 and 0.17 mM magnesium DL-isocitrate), a number of differences in responses to the addition of EGTA and Ca^{2+} were noted under these conditions, compared to experiments (Figs. 1 to 3) where substrate concentrations were maintained with high magnesium: isocitrate ratios: (i) Without ADP, EGTA resulted in a relatively small inhibition of activity although the inhibition was still appreciable, especially at the lower concentration of magnesium isocitrate (e.g., about 30% inhibition was obtained at 100 μM EGTA with 0.065 mM magnesium DL-isocitrate) (Fig. 4). (ii) With ADP, EGTA markedly inhibited activity. At 0.065 mM magnesium DL-isocitrate, the activity retained declined to a relatively constant rate between 20 and 1000 μM EGTA (53 to 44%), compared to the control not containing EGTA (Fig. 4); the decrease in velocity was less with 0.17 mM magnesium DL-isocitrate. However, at either substrate concentration the activity even with 5000 μM EGTA was still strikingly higher with than without ADP (compare A with B and C with D in Fig. 4). Since the level of endogenous total calcium was about 3 μM in these experiments, it became apparent that Ca^{2+} was not absolutely required for activation of the enzyme by ADP. While this was readily apparent when high isocitrate: magnesium ratios were used to maintain substrate concentration (Fig. 4) the same trend could be noted, albeit to a much smaller extent, at high magnesium: isocitrate ratios (e.g., compare the activities in the presence of 100 μM EGTA and increasing ADP in curve B of Fig. 2). (iii) Without ADP, the addition of calcium had little effect on activity in the absence of EGTA; however, with EGTA calcium substantially inhibited activity (Fig. 5). The latter was in contrast to results obtained when high magnesium: isocitrate ratios were used (Fig. 3). (iv) Inhibition by combinations of calcium and EGTA was also observed in the presence of ADP. When compared on the basis of free Ca^{2+} concentrations (Fig. 6), the velocities were nearly the same up to about 1 μM Ca^{2+} with and without EGTA. However, above 1 μM free Ca^{2+} the rates were appreciably larger in the absence than in presence of EGTA.

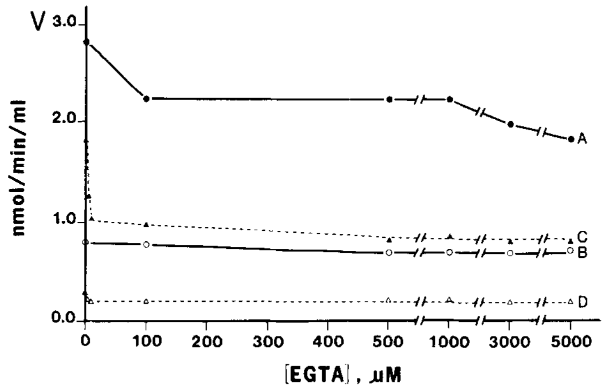

FIG. 4. Varying EGTA with and without ADP at high isocitrate:magnesium ratios. Reaction mixtures contained 52 mM DL-isocitrate and 0.070 or 0.18 mM $MgSO_4$. The endogenous calcium content was 3 μM. EGTA was varied between 0 and 5000 μM under the following conditions: (A) magnesium isocitrate was about 0.17 mM and ADP was 0.50 mM; (B) magnesium isocitrate was about 0.17 mM; (C) magnesium isocitrate was about 0.065 mM and ADP was 0.50 mM; (D) magnesium isocitrate was about 0.065 mM.

It is apparent from the above observations that EGTA can decrease enzyme activity in a manner which cannot be correlated simply with reduction of the level of activating Ca^{2+} at the allosteric binding site for ADP. The inhibition in the absence of ADP occurred at low concentrations of EGTA, it was not complete, and the activity retained remained constant over a wide range of EGTA concentrations when the level of free Mg^{2+} was high (Fig. 1). A similar pattern was observed for inhibition by Ca^{2+} when EGTA was present and free Mg^{2+} was low (Fig. 5). It may be that these responses to EGTA or the combination of EGTA and calcium, respectively, are due to conformational changes of the enzyme to a less active form(s), perhaps similar to changes caused by the effects of temperature and enzyme concentration on NAD-isocitrate dehydrogenase from bovine heart (23) and porcine heart (24). The inhibition by calcium reported for NAD-isocitrate dehydrogenase from a number of species and tissues (10, 11, 13) may be mediated by EGTA in a manner similar to that noted here. In any case, all

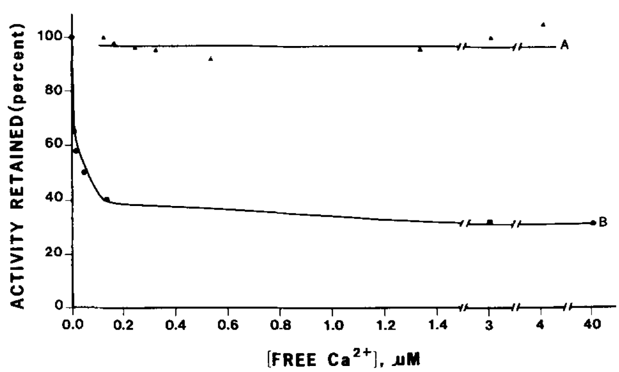

FIG. 5. Inhibition of activity by Ca^{2+} with EGTA at high isocitrate:magnesium ratio. The media contained 52 mM DL-isocitrate and 0.18 mM $MgSO_4$ maintaining magnesium isocitrate concentration at 0.172–0.174 mM. The media contained 3 μM calcium without added $CaCl_2$. (A) EGTA absent; from 0 to 100 μM $CaCl_2$ was added; (B) EGTA (5.0 mM) was present; from 0 to 6 mM $CaCl_2$ was added. The concentrations of free Ca^{2+} were calculated from the stability constants. The samples without added Ca^{2+} were taken to have 100% activity.

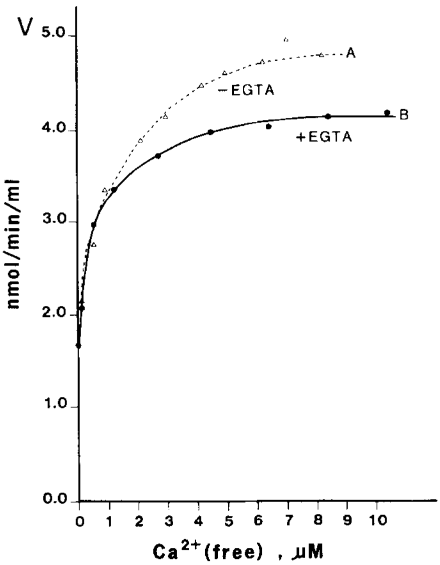

FIG. 6. The effect of Ca^{2+} in presence of ADP at high isocitrate:magnesium ratio. The media contained 52 mM DL-isocitrate, 0.18 mM $MgSO_4$, and 0.50 mM ADP. (A) Without EGTA, the concentration of Ca^{2+} was varied by the addition of up to 200 µM $CaCl_2$; (B) with EGTA (1.0 mM), Ca^{2+} was varied by the addition of 500 to 1250 µM $CaCl_2$.

studies on the effects of Ca^{2+} on this enzyme, so far, have utilized EGTA to regulate the concentration of free Ca^{2+} (10–14).

Kinetic constants. In order to avoid the artifacts which could be introduced by the use of EGTA in estimating the effects of Ca^{2+} on kinetic constants of the enzyme, other approaches for regulating Ca^{2+} levels were explored. The stability constants of the complexes of 1,3-diamino-2-hydroxypropane-*N*,*N*,*N*′,*N*′-tetraacetate (DHPTA) with Ca^{2+} and Mg^{2+} favor the chelation of Ca^{2+} over Mg^{2+} by a ratio of 68 at pH 7.4. In addition, the stability constant for Ca^{2+} with DHPTA is smaller by several orders of magnitude than that with EGTA. Nevertheless, DHPTA and EGTA had similar effects on bovine heart NAD-isocitrate dehydrogenase (not shown). Experiments on the determination of kinetic constants were therefore done under conditions where levels of endogenous Ca^{2+} were minimized by careful attention to quality of reagents and other sources of Ca^{2+} contamination (e.g., all solutions were prepared and stored in Teflon vessels). The concentrations of calcium determined by atomic adsorption were used for calculations of the kinetic parameters. High total isocitrate: magnesium ratios were used to maintain substrate concentration. This had the advantage that the inhibition by free Mg^{2+} was minimal, ADP was mainly in the active nonchelated form (3), and isocitrate acted as a buffer to maintain free Ca^{2+} concentration. EGTA was absent from all reaction mixtures, except for the series of determinations of velocities where EGTA was added to reduce free Ca^{2+} to zero (v_0). The latter seemed justified since under conditions where total Ca^{2+} was low inhibition by EGTA was minimal or absent (Fig. 6). Velocities were determined at different fixed levels of Ca^{2+} with ADP varied. As shown in Fig. 7, velocity increased with ADP when the level of free Ca^{2+} was zero (the concentration of endogenous total Ca^{2+} was 3 µM and EGTA was 100 µM); however, the rates at each concentration of ADP increased progressively with increasing levels of free Ca^{2+}. A replot of the data of Fig. 7 as the reciprocals of ADP vs the reciprocals of (v-v_0), where values of v_0 were the velocities

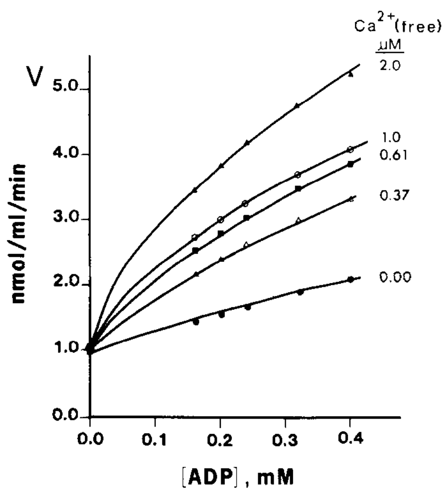

FIG. 7. Velocity with varying ADP and Ca^{2+}. Magnesium DL-isocitrate was maintained at 0.170–0.174 mM and free Mg^{2+} at 6.3–6.4 µM with 0.18 mM $MgSO_4$ and 52 mM DL-isocitrate. When present, ADP was varied from 0.16–0.40 mM. For experiments containing "zero" free Ca^{2+}, 100 µM EGTA was added (the endogenous total calcium content of the reaction mixture was 2.3 µM). The fixed levels of free Ca^{2+} were adjusted to 0.37, 0.61, 1.0, and 2.0 µM by the addition of $CaCl_2$ without EGTA.

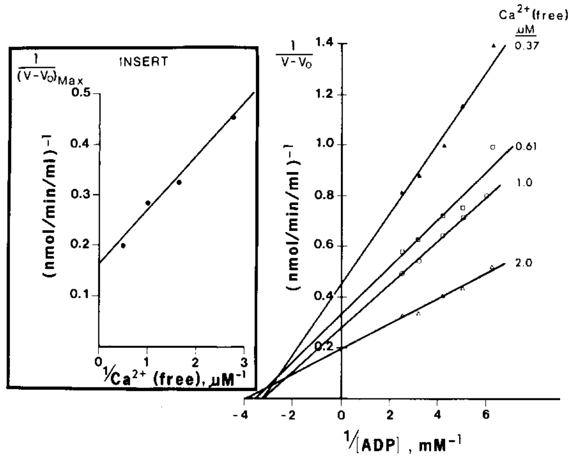

FIG. 8. Double reciprocal plot of velocity vs ADP as a function of free Ca^{2+}. The data from Fig. 7 were plotted as reciprocals of v-v_0 vs ADP; where v was the velocity at each level of ADP with varying fixed Ca^{2+} and v_0 was the velocity at "zero" free Ca^{2+} (100 μM EGTA present) at each level of ADP. The lines were fitted to the data for each level of Ca^{2+} by the computer program of Cleland (21) for a rectangular hyperbola not intersecting at the origin. *Insert:* Secondary plot of ordinate intercepts ($1/(v$-$v_0)$ vs $1/Ca^{2+}$. The line was fitted to the data by the method of least squares.

at each level of ADP when free Ca^{2+} was zero (EGTA present) and v were the velocities at each level of ADP at different fixed concentrations of free Ca^{2+} (EGTA absent), gave a series of straight lines with markedly different intercepts at the vertical axis (Fig. 8). Replots of ordinate intercepts vs the reciprocals of free Ca^{2+} were linear (insert, Fig. 8), and K_m for free Ca^{2+} was calculated from the abscissa intercept. The constants from three experiments of the type shown in Figs. 7 and 8 are summarized in Table I. Values of K_m (app) for ADP with varying Ca^{2+} for each of the experiments were in a relatively narrow range and within the standard deviation of the mean, suggesting that Ca^{2+} had little or no effect on K_m for ADP. The concentration of magnesium DL-isocitrate was larger in Experiment No. 3 (0.25 mM) than in Experiments Nos. 1 and 2 (0.17 mM) (Table I). The differences in values of K_m for ADP between these experiments are consistent with the previous observation that there is a reciprocal effect of magnesium isocitrate and ADP concentrations on their respective K_m values (3).

TABLE I

KINETIC CONSTANTS FOR INTERACTION OF NAD-ISOCITRATE DEHYDROGENASE WITH Ca^{2+} AND ADP

Experiment No.	Fixed levels of free Ca^{2+} (μM)	Range of K_m (app) for ADP with varying Ca^{2+} (mM)	K_m (app) for ADP (mean ± SD) (mM)	K_m for Ca^{2+} (μM)
1	0.37, 0.61, 1.0, 2.0	0.247–0.308	0.275 ± 0.030	0.65 (0.940)[a]
2	0.37, 0.61, 1.0	0.208–0.272	0.240 ± 0.032	0.61 (0.999)
3	0.25, 0.74, 1.2, 2.1, 4.1	0.082–0.095	0.090 ± 0.007	0.30 (0.999)

Note. The conditions of incubation, compositions of reaction mixtures, and calculations are described in Figs. 7 and 8 for Experiments 1 and 2. In Experiment No. 3, the concentrations of magnesium isocitrate and free Mg^{2+} were maintained at 0.250 and 0.020 mM, respectively, with 24 mM DL-isocitrate and with $MgSO_4$ varied from 0.27–0.30 mM; ADP was varied from 0.16–0.65 mM.

[a] The number in parentheses is the coefficient of correlation r^2 for the fit of the line by the method of least squares to the replot of ordinate intercepts $(1/(v$-$v_0)$max) vs $1/Ca^{2+}$.

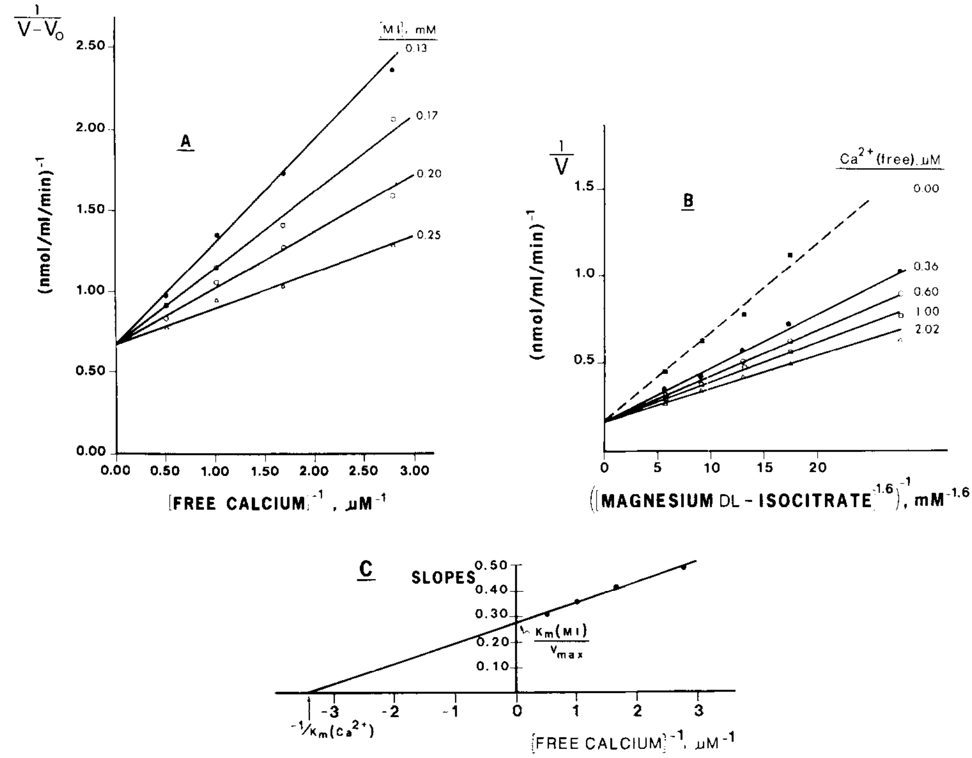

FIG. 9. Velocity with varying Ca^{2+} and magnesium isocitrate. The media contained 0.30 mM ADP, 52 mM DL-isocitrate, and the endogenous Ca^{2+} content was 3.0 μM. The concentrations of magnesium DL-isocitrate were varied between 0.13 and 0.25 mM by the addition of from 0.13 to 0.26 mM $MgSO_4$. For experiments where velocity was measured at "zero" free Ca^{2+} (v_0), 100 μM EGTA was added. The levels of free Ca^{2+} were maintained between 0.36 and 2.0 μM as described in Fig. 7. (A) Double reciprocal plot of velocity (v-v_0) vs free Ca^{2+} as a function of magnesium DL-isocitrate (MI). The lines were plotted as described in Fig. 8. (B) Double reciprocal plot of velocity vs magnesium DL-isocitrate as a function of free Ca^{2+} concentration. The exponent for magnesium isocitrate concentration is the Hill slope ($n = 1.6$) determined from a Hill plot (not shown). The lines were fitted to the data for each level of Ca^{2+} by the computer program of Cleland (21) for a rectangular hyperbola. At zero Ca^{2+} the K_m for magnesium DL-isocitrate was 0.35 mM. (C) Secondary plot of data from B. The line was fitted to the data by the method of least squares ($r^2 = 0.989$). See the text for values of calculated constants.

The stimulation of rate by Ca^{2+} in the presence of ADP was mainly due to the increase of apparent V (Fig. 8).

The values of K_m for free Ca^{2+} obtained under the conditions shown in Table I appeared to be dependent on the concentration of magnesium isocitrate. The latter is supported by experiments in which ADP was kept constant (0.30 mM) and magnesium isocitrate and Ca^{2+} were varied as shown in Fig. 9. K_m (app) for free Ca^{2+} decreased with increasing fixed levels of magnesium isocitrate (Fig. 9A) and K_m (app) for magnesium isocitrate declined with increasing fixed levels of Ca^{2+} (Fig. 9B). From a secondary plot of the data in Fig. 9B, values of limiting K_m for free Ca^{2+} and magnesium DL-isocitrate were estimated to be 0.29 μM and 0.19 mM, respectively (Fig. 9C). Values of maximal velocity were the same when either the fixed levels of magnesium isocitrate (Fig. 9A) or free Ca^{2+} (Fig. 9B) were varied. When ADP was omitted under conditions which were otherwise identical with those shown in Fig. 9, Ca^{2+} had no effect on velocity, the K_m for magnesium DL-isocitrate was 0.48 mM, and the Hill slope increased from $n = 1.6$ with ADP (Fig. 9B) to $n = 2.0$ without ADP (not shown); consistent with previous reports

(3), V was the same with ADP (Fig. 9B) and without ADP (not shown).

These results indicate that: (i) Ca^{2+} does not activate in the absence of ADP (Fig. 7). (ii) Ca^{2+} facilitates the lowering of K_m for magnesium isocitrate by ADP (Fig. 9); however, Ca^{2+} is not absolutely required for activation by ADP (Figs. 2 and 7). (iii) The addition of Ca^{2+} to the enzyme may occur subsequent to binding of ADP, since Ca^{2+} increased velocity but not K_m (app) for ADP with subsaturating concentrations of magnesium isocitrate (Fig. 8 and Table I).

Effects of Potential Calcium Antagonists or Agonists

Certain antipsychotic drugs containing a phenothiazine ring structure have been reported to interfere with specific calcium-binding processes. Thus, trifluoperazine was found to prevent the activation of phosphorylase kinase by the calcium-binding protein calmodulin (25, 26); chlorpromazine prevented damage by Ca^{2+} to mitochondria from ischemic rat livers (27).

The effects of chlorpromazine and trifluoperazine on NAD-isocitrate dehydrogenase were tested in the presence and absence of ADP with and without added Ca^{2+} (Table II). Soderling and Srivastava (26) reported that the calmodulin-stimulated phosphorylation of glycogen synthetase was blocked by 0.1 mM trifluoperazine. This concentration of trifluoperazine or chlorpromazine did not inhibit NAD-isocitrate dehydrogenase either with or without ADP or Ca^{2+} (Table II). The concentration ratio of magnesium to DL-isocitrate was 1.7:1 in Table II. The drugs also did not affect enzyme activity when substrate concentration was maintained with an isocitrate: magnesium ratio of 90:1 (not shown).

La^{3+} and a number of lanthanide ions have been found to be antagonists of Ca^{2+} fluxes in a variety of tissues (for review see Weiss (28)), and La^{3+} inhibited the uptake or binding of Ca^{2+} by rat liver mitochondria (29, 30). Binding of lanthanide ions at the calcium-binding sites of several enzymes has been reported. For example, Darnall and Birnbaum (31) observed that the conversion of trypsinogen to trypsin was activated more effectively by neodymium ion than by Ca^{2+}, and that certain lanthanide ions acted as calcium agonists for α-amylase (32). The effect of substitution of lanthanide ions for calcium on the calcium-dependent activation of factor X by Russel's viper venom has been investigated (33), and the binding of a series of lanthanide ions to the calcium-binding site of trypsin has been studied systematically (34, 35).

TABLE II

EFFECT OF CHLORPROMAZINE AND TRIFLUOPERAZINE

		Drug		
		Chlorpromazine		Trifluo-perazine
Additions	None	50 μM	100 μM	100 μM
None	0.64	0.64	0.58	0.69
ADP	1.43	1.50	1.60	1.47
$CaCl_2$	0.74	0.73	0.73	0.76
ADP *plus* $CaCl_2$	3.84	4.10	4.20	4.05

Note. The reaction mixtures contained 1.0 mM $MgSO_4$ and 0.60 mM DL-isocitrate. When present, the concentration of ADP was 0.20 mM and $CaCl_2$ was 0.10 mM. The concentrations of drugs were as specified. Velocities are reported as nmol/min/ml.

La^{3+} was found to be a potent inhibitor of NAD-isocitrate dehydrogenase; however, inhibition was obtained both with and without ADP (Fig. 10). In the absence of ADP, Ca^{2+} did not reverse the inhibition by La^{3+}. In the presence of ADP, an increase in activity was obtained with or without La^{3+} as the concentration of added Ca^{2+} increased. However, calcium was not competitive with La^{3+} since the maximal velocity (above 50 μM Ca^{2+}) with 5 μM La^{3+} was markedly lower than without La^{3+} (Fig. 10).

The inhibition by La^{3+} was competitive with respect to magnesium isocitrate under conditions where free Mg^{2+} concentration was kept constant (Fig. 11). The same type of inhibition was found with a number of other lanthanide ions (Table III). The values of K_{is} were rather similar for La^{3+}, Yb^{3+}, Gd^{3+}, Eu^{3+}, Tb^{3+} and Er^{3+}, ranging from 1.8 to 3.1 μM. The similarity in binding affinities of the lanthanides for the isocitrate dehydrogenase contrasts to the large dif-

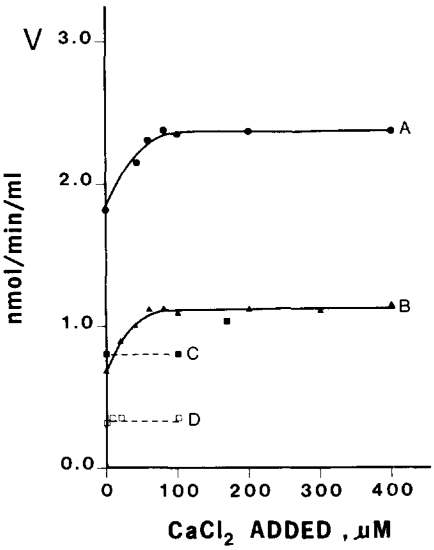

FIG. 10. Effect of added calcium on inhibition by $LaCl_3$ with and without ADP. All reaction mixtures contained 167 mM Na-Hepes, 0.5 mM DL-isocitrate, 1.0 mM $MgSO_4$, and 0.5 mM NAD^+. (A) ADP (0.5 mM) present; (B) ADP (0.5 mM) and $LaCl_3$ (5 μM) present; (C) without ADP; (D) with $LaCl_3$ (5 μM) and without ADP.

TABLE III

INHIBITION OF NAD-SPECIFIC ISOCITRATE DEHYDROGENASE BY LANTHANIDES

Inhibitor	Inhibitor concn range (μM)	Inhibition constant, K_{is}[a] (μM)	K_m for magnesium isocitrate[a] (mM)
$LaCl_3$	3–10	1.94 ± 0.17	0.22 ± 0.02
$Yb(NO_3)_3$	3–10	1.81 ± 0.12	0.22 ± 0.01
$Gd(NO_3)_3$	3–10	2.95 ± 0.21	0.22 ± 0.02
$EuCl_3$	3–10	3.10 ± 0.20	0.20 ± 0.02
$TbCl_3$	5–10	2.42 ± 0.21	0.25 ± 0.04
$ErCl_3$	2–5	2.81 ± 0.37	0.20 ± 0.13

Note. All assay media contained 167 mM Na-Hepes at pH 7.4, 0.5 mM NAD^+, 0.67 mM ADP, and enzyme. $MgSO_4$ was varied from 0.94 to 1.15 mM and DL-isocitrate from 0.30 to 1.00 mM, to vary magnesium DL-isocitrate from 0.092 to 0.306 mM while free Mg^{2+} remained constant at 0.85 mM. The concentrations of lanthanide salts were fixed at the levels indicated. The reactions were started with NAD^+ after the otherwise complete reaction mixture containing enzyme had been incubated for 1 min at 25°C.

[a] Mean ± SE.

ferences in binding constants of these ions for the metal binding site of porcine trypsin; these have been correlated with differences in the ionic radii of the cations (35). The lanthanide ions are very effective inhibitors of NAD-isocitrate dehydrogenase; the values of inhibition constants (1.8 to 3.1 μM) were about two orders of magnitude lower than K_m for magnesium DL-isocitrate obtained under the same conditions (Table III). It remains to be established, however, whether the interaction with NAD-isocitrate dehydrogenase occurs with the lanthanide ions per se or, possibly, with complexes of lanthanides with isocitrate.

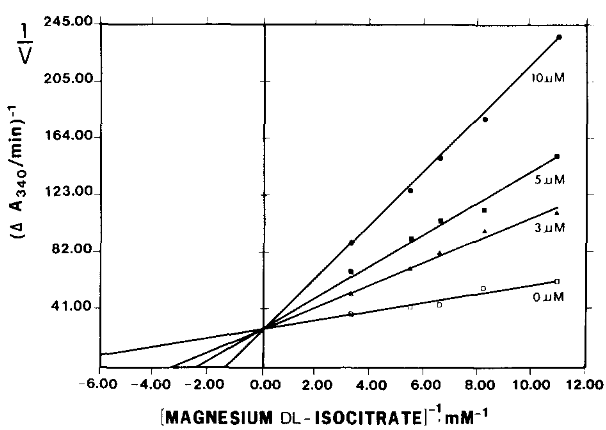

FIG. 11. Inhibition of NAD-specific isocitrate dehydrogenase by $LaCl_3$ as a function of magnesium isocitrate concentration. The concentrations of $LaCl_3$ were fixed at 0, 3, 5, and 10 μM, as shown at each line. The concentration of magnesium isocitrate was varied and free Mg^{2+} was maintained at constant concentration as described in Table III.

REFERENCES

1. CHEN, R. F., AND PLAUT, G. W. E. (1963) *Biochemistry* **2**, 1023–1032.
2. GOEBELL, H., AND KLINGENBERG, M. (1964) *Biochem. Z.* **340**, 441–464.
3. PLAUT, G. W. E., SCHRAMM, V. L., AND AOGAICHI, T. (1974) *J. Biol. Chem.* **249**, 1848–1856.
4. CHEN, R. F., BROWN, D. M., AND PLAUT, G. W. E. (1964) *Biochemistry* **3**, 552–559.
5. PLAUT, G. W. E., AND AOGAICHI, T. (1968) *J. Biol. Chem.* **243**, 5572–5583.
6. STEIN, A. M., KIRKMAN, S. K., AND STEIN, J. H. (1967) *Biochemistry* **6**, 3197–3203.
7. KORNBERG, A., AND PRICER, W. E., JR. (1951) *J. Biol. Chem.* **189**, 123–136.
8. PLAUT, G. W. E., AND SUNG, S.-C. (1954) *J. Biol. Chem.* **207**, 305–314.
9. PLAUT, G. W. E. (1970) *in* Current Topics in Cellular Regulation (Horecker, B. L., and Stadtman, E. R., eds.), Vol. 2, pp. 1–26, Academic Press, New York.
10. VAUGHAN, H., AND NEWSHOLME, E. A. (1969) *FEBS Lett.* **5**, 124–126.
11. ZAMMIT, V. A., AND NEWSHOLME, E. A. (1976) *Biochem. J.* **154**, 677–687.
12. BULOS, B. A., AND SACKTOR, B. (1979) *Anal. Biochem.* **95**, 62–72.
13. BULOS, B., AND THOMAS, J. (1979) *Fed. Proc.* **38**, 505.
14. DENTON, R. M., RICHARDS, D. A., AND CHIN, J. G. (1978) *Biochem. J.* **176**, 899–906.
15. ENGLAND, P. J., DENTON, R. M., AND RANDLE, P. J. (1967) *Biochem. J.* **105**, 32C–33C.
16. GIORGIO, N. A., JR., YIP, A. T., FLEMING, J., AND PLAUT, G. W. E. (1970) *J. Biol. Chem.* **245**, 5469–5477.
17. PLAUT, G. W. E. (1969) *in* Methods in Enzymology (Lowenstein, J. M., ed.), Vol. 13, pp. 34–42, Academic Press, New York/London.
18. GRZYBOWSKI, A. K., TATE, S. S., AND DATTA, S. P. (1970) *J. Chem. Soc. Sec. A*, 241–245.
19. SILLEN, L. G., AND MARTELL, A. E., (1964) Stability Constants of Metal-Ion Complexes, The Chem. Soc., London.
20. FELDMAN, H., RODBARD, D., AND LEVINE, D. (1972) *Anal. Biochem.* **45**, 530–556.
21. CLELAND, W. W. (1963) *Nature (London)* **198**, 463–465.
22. FAN, C. C., STEGEMAN, W. J., AND PLAUT, G. W. E. (1977) *Arch. Biochem. Biophys.* **184**, 125–134.
23. FAN, C. C., LIN, J. -P. F., AND PLAUT, G. W. E. (1975) *J. Biol. Chem.* **250**, 2022–2027.
24. SHEN, W.-C., MAUCK, L., AND COLMAN, R. F. (1974) *J. Biol. Chem.* **249**, 7942–7949.
25. COHEN, P. (1979) *Fed. Proc.* **39**, 788 (Abstr. 2948).
26. SODERLING, T. R., AND SRIVASTAVA, A. K. (1979) *Fed. Proc.* **39**, 788 (Abstr. 2949).
27. MITTNACHT, S. JR., SHERMAN, S. C., AND FARBER, J. L. (1979) *J. Biol. Chem.* **254**, 9871–9878.
28. WEISS, G. B. (1974) *Annu. Rev. Pharmacol.* **14**, 343–354.
29. MELA, L. (1968) *Arch. Biochem. Biophys.* **123**, 286–293.
30. LEHNINGER, A. L., AND CARAFOLI, E. (1971) *Arch. Biochem. Biophys.* **143**, 506–515.
31. DARNALL, D. W., AND BIRNBAUM, E. R. (1970) *J. Biol. Chem.* **245**, 6484–6486.
32. DARNALL, D. W., AND BIRNBAUM, E. R. (1973) *Biochemistry* **12**, 348.
33. FURIE, B. C., AND FURIE, B. (1975) *J. Biol. Chem.* **250**, 601–608.
34. EPSTEIN, M., LEVITZKI, A., AND REUBEN, J. (1974) *Biochemistry* **13**, 1777–1782.
35. EPSTEIN, M., REUBEN, J., AND LEVITZKI, A. (1977) *Biochemistry* **16**, 2449–2457.